DAIICT

IT629 INTRODUCTION TO ALGORITHMS

FIRST IN-SEMESTER EXAMINATION

ID NUMBER: 202312081 NAME: Vora Nidhi Maheshbhai	SET A

NOTES:

1. Total MARKS: 10 x 2 = 20 TIME allowed: 1 HOUR.
2. Write your answers in the boxes provided below.

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	ANSWER
Q-1 2	Summation : 1830
Q-2	Smallest Value of k : 2 $O(n^2)$
Q-3 2	$T(n) = \Theta(n^2)$
Q-4 2	loop invariant - $j = 2 \ldots\ldots$ length[A]
Q-5 2	insertion sort Best case = $O(n)$ Quick sort Worst case = $\Theta(n^2)$
Q-6	Heap Sort Time Complexity = $\Theta(n \log n)$
Q-7	leaf node : 20
Q-8 2	If G is connected Graph $\lvert E \rvert = \lvert V \rvert - 1$ If you delete any vertices of G then two disconnected node forms.
Q-9 2	key[y] ≤ key[z]
Q-10	upper limit $h = \sum_{h=0}^{\infty} (h/2^h)$ Ans = 2

Q-1. Calculate the value of the following summation:

$\Sigma_{k=21..40}$ 3*k

Q-2. What is the smallest integer value of k such that $100n^2(\lg n)$ is in $O(n^k)$?

Q-3. Using the Master Theorem, state the solution to the following recurrence

$T(n) = 4T(n/2) + 2n$

Q-4. Refer to the function **INSERTION-SORT (A)** given below. State the loop invariant of the outer **for** loop

```
INSERTION-SORT(A)
for j = 2 to length[A]
  key = A[j];
  i = j-1;
  while i > 0 and A[i] > key
    A[i+1] = A[i];
    i = i-1;
  A[i+1] = key;
```

Q-5. State (a) the best case running time of insertion sort, and (b) the worst case running time of Quicksort.

Q-6. (a) Name one sort algorithm which is NOT based on comparisons, and (b) state the running time of this algorithm in Θ notation.

Q-7. A binary heap has 41 nodes. How many of these nodes are leaf nodes?

Q-8. Let G = (V,E) be an undirected graph. State any one condition that G must satisfy to become a tree. Recall that we have seen five equivalent properties of a tree.

Q-9. T is a binary search tree, and X is an internal node of T. Y is a node in the left subtree of X, and Z is a node in the right subtree of X. If values of key[Y] and key[Z] are compared, what is the relationship between these two values?

Q-10. A B-tree is stored as a file on secondary storage, with 2×10^7 records. The node degree of this data structure is defined by minimum degree t = 100. Determine the upper limit h of the B-tree height.

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FIRST IN-SEMESTER EXAMINATION

ID NUMBER: 202312019

NAME: Shah Helvi Jignubhai

SET B

16/20

NOTES:

1. Total MARKS: 10 x 2 = 20 TIME allowed: 1 HOUR
2. Write your answers in the boxes provided below.

	ANSWER
Q-1	22 leaf node ($\lfloor \frac{43}{2} \rfloor + 1$) ✓
Q-2	There should be no cycle in a graph i.e the node should not be traversed to itself using the edges. ✓
Q-3	key[Y] < key[Z] ✓
Q-4	2×10^4 ✗
Q-5	1065 ✓
Q-6	K = 4 ✓
Q-7	$n^2 \log n$ ✓
Q-8	A[1 ⋯ j-1] elements are always sorted. ✓
Q-9	(a) Worst case running time of Quick sort = $O(n^2)$ (b) Best case running time of Insertion sort = $O(n)$ ✓
Q-10	(a) Heap sort (b) $\Theta(n \log n)$ ✗

Q-1. A binary heap has 43 nodes. How many of these nodes are leaf nodes?

Q-2. Let G = (V,E) be an undirected graph. State any one condition that G must satisfy to become a tree. Recall that we have seen five equivalent properties of a tree.

Q-3. T is a binary search tree, and X is an internal node of T. Y is a node in the left subtree of X, and Z is a node in the right subtree of X. If values of key[Y] and key[Z] are compared, what is the relationship between these two values?

Q-4. A B-tree is stored as a file on secondary storage, with 2×10^{9} records. The node degree of this data structure is defined by minimum degree $t = 1000$. Determine the upper limit h of the B-tree height.

Q-5. Calculate the value of the following summation:

$$\sum_{k=31..40} 3*k$$

Q-6. What is the smallest integer value of k such that $10n^3(\lg n)$ is in $O(n^k)$?

Q-7. Using the Master Theorem, state the solution to the following recurrence:

$$T(n) = 4T(n/2) + n^2$$

Q-8. Refer to the function **INSERTION-SORT(A)** given below. State the loop invariant of the outer **for** loop.

```
INSERTION-SORT(A)
for j = 2 to length[A]
  key = A[j];
  i = j-1;
  while i > 0 and A[i] > key
    A[i+1] = A[i];
    i = i-1;
  A[i+1] = key;
```

Q-9. State (a) the worst case running time of Quicksort, and (b) the best case running time of insertion sort.

Q-10. (a) Name one sort algorithm which is NOT based on comparisons, and (b) state the running time of this algorithm in Θ notation.